

Computer Networking Homework #1.1

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1. What is the analog bandwidth of a signal?

Ans: Use Fourier Analysis to decompose an analog signal, we can get various sine waves. And we subtract the lowest frequency from the highest one, we can get the bandwidth

2. Suppose an engineer increases the number of possible signal levels of a digital signal from 2 to 8. How many more bits can be sent in the same amount of time? Explain.

We can use Nyquist's Theorem to solve this:

$$Data\ Rate_{Levels=2} = 2 \times B \times \log_2(2) = 1B$$

$$Data\ Rate_{Levels=8} = 2 \times B \times \log_2(8) = 3B$$

$$3 - 1 = 2$$

ANS: We can send 2 more bit in the same time

3. If the maximum frequency audible to a human ear is 20,000 Hz, at what rate must the analog signal from a microphone be sampled when converting it to digital?

By Nyquist Theorem: $Sample\ Rate = 2 \times f_{max}$

$$20000 \times 2 = 40000$$

ANS: 40000

4. Identify the three major steps to convert an analog signal to a digital signal.

ANS: sampling, quantization, encoding

5. If a system has an average signal power level of 100, an average noise level of 33.33, and a bandwidth of 100 MHz, what is the effective limit on channel capacity?

By the Shannon's Theorem: $Channel\ Capacity = B \times \log_2(1 + S/N)$

$$100MHz \times \log_2(1 + \frac{100}{33.33}) \approx 200MHz$$

ANS: 200Mhz

6. Two major modulation techniques for an analog signal are Amplitude Modulation (AM) and Frequency Modulation (FM). Explain the differences between these two modulation schemes.

ANS:

AM: 使用振幅表示訊號的大小，原訊號振幅越大，調變後的振幅也就越大，且反之亦然

FM: 使用頻率表示訊號的大小，原訊號振幅越大，調變後的頻率就越大，且反之亦然